ANLI Round 2

Multi-Class Natural Language Inference Classification

End-to-end ML pipeline: EDA, Baseline, Transformer Fine-tuning, Docker Deployment

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What is Natural Language Inference?

Premise:
"Idris Sultan (born January 1993) is a Tanzanian Actor and comedian."

Three possible relationships:
Entailment: "Born in Tanzania"
Neutral: "Won multiple awards"
Contradiction: "Born in 1990"

Why ANLI is Different

ANLI (Adversarial NLI) was collected using a human-and-model-in-the-loop process. Human annotators specifically wrote hypotheses to fool RoBERTa — making Round 2 far harder than SNLI or MNLI. Surface-level features like word overlap provide zero predictive signal.

Dataset Overview — ANLI Round 2

train_r2	dev_r2	test_r2
45,460	**1,000**	**1,000**

Premise Length	54 words avg (33-76)
Hypothesis Length	10 words avg (3-65)
Token Length	83 avg, max 153 (100% fit in 256)
Labels (Train)	Neutral 46.1% | Entailment 31.8% | Contradiction 22.1%
Word Overlap	Nearly identical across labels (~0.10-0.14 Jaccard)

Pipeline Overview

Data Loading & EDA

Load R2 splits, analyze label distribution, text lengths, word overlap

Preprocessing

DeBERTa tokenizer
[CLS] premise [SEP] hypothesis [SEP]
max_length=256

Baseline Model

TF-IDF + Logistic Regression
Establish floor
(32.3% accuracy)

DeBERTa-v3 Fine-tuning

184M params, 3 epochs
lr=2e-5, batch=16
fp32 on T4 GPU

Evaluation & Error Analysis

Confusion matrices
confidence analysis
error categorization

Model Selection — Why DeBERTa-v3-base?

Disentangled Attention

Separately encodes content (word meaning) and position (word location), computing attention from both signals independently. Better than BERT/RoBERTa which combine them.

Replaced Token Detection

Pre-trained with RTD instead of Masked Language Modeling. The model learns from every token (not just 15% masked), making it more sample-efficient.

Adversarial Robustness

ANLI was built to fool RoBERTa/BERT. DeBERTa's different architecture is less susceptible to the adversarial patterns in this dataset.

Proven ANLI Performance

DeBERTa-v3-base outperforms most large models on the ANLI benchmark. The base model (184M params) fits comfortably on a T4 GPU.

Training Configuration & Progress

Hyperparameters

Model	`microsoft/deberta-v3-base`
Parameters	`184,424,451`
Epochs	`3`
Batch Size	`16`
Learning Rate	`2e-5`
Warmup Steps	`300`
Weight Decay	`0.01`
Max Length	`256 tokens`
Precision	`FP32 (T4 GPU)`

Training Progress

Epoch	Train Loss	Val Loss	Accuracy
1	0.858	1.299	37.0%
2	0.819	1.316	37.2%
3	**0.777**	**1.364**	**40.5%**

Training loss decreased steadily (1.1 → 0.7). Validation loss increased slightly after epoch 2, indicating early signs of overfitting — expected for adversarial data with limited training examples.

Results — Baseline vs DeBERTa

32.3%
TF-IDF + LR
(Baseline)

40.5%
DeBERTa-v3
(Dev)

39.3%
DeBERTa-v3
(Test)

+25%
Relative
Improvement

Per-Class Performance (Dev Set)

Class	Precision	Recall	F1-Score	Support
Entailment	0.360	0.569	0.441	334
Neutral	0.462	0.423	0.442	333
Contradiction	0.443	0.222	0.296	333

Error Analysis

Error Distribution (595 / 1,000 = 59.5%)

Top confusion pairs (true → predicted):

Contradiction → Entailment: 188 errors
Neutral → Entailment: 150 errors
Entailment → Neutral: 93 errors

Key insight: The model has a strong entailment bias — it predicts entailment for 528/1000 examples. This is driven by the training set imbalance.

Failure Categories

Negation & Contradiction
Model fails to detect when hypothesis contradicts premise through negation or date conflicts

Numerical Reasoning
Cannot perform arithmetic or compare dates/numbers accurately

Multi-hop Inference
Struggles when connecting facts requires 2+ reasoning steps

Longer Premises
Accuracy drops from 41.5% (30-50 words) to 29.4% (70-100 words)

World Knowledge
Fails when inference requires knowledge not in the premise text

Docker Deployment

```
# Build & Run
docker build -t anli-r2-classifier .
docker run -p 8000:8000 anli-r2-
classifier

# Test endpoint
curl -X POST localhost:8000/predict \
  -H "Content-Type: application/json" \
  -d '{"premise": "...",
       "hypothesis": "..."}'

# Response
{"prediction": "entailment",
 "confidence": 0.82,
 "probabilities": {...}}
```

Architecture

- FastAPI + Uvicorn server
- POST /predict endpoint
- POST /predict/batch for bulk inference
- GET /health for monitoring
- Swagger docs at /docs

Container Specs

- Base: python:3.10-slim
- Model: ~370MB safetensors checkpoint
- Memory: ~4GB at runtime
- Health checks every 30s

Future Improvements

Data Augmentation
High Impact

Combine SNLI + MNLI + FEVER-NLI with ANLI R2 — models trained this way achieve ~70% accuracy on ANLI

Larger Model
Medium

DeBERTa-v3-large (304M params) for ~5-8% accuracy improvement

Class Weighting
Medium

Address training imbalance (contradiction is underrepresented at 22%)

Ensemble
Medium

Combine predictions from multiple models/checkpoints

Hyperparameter Search
Low-Medium

Systematic tuning of learning rate, warmup, batch size

Questions?